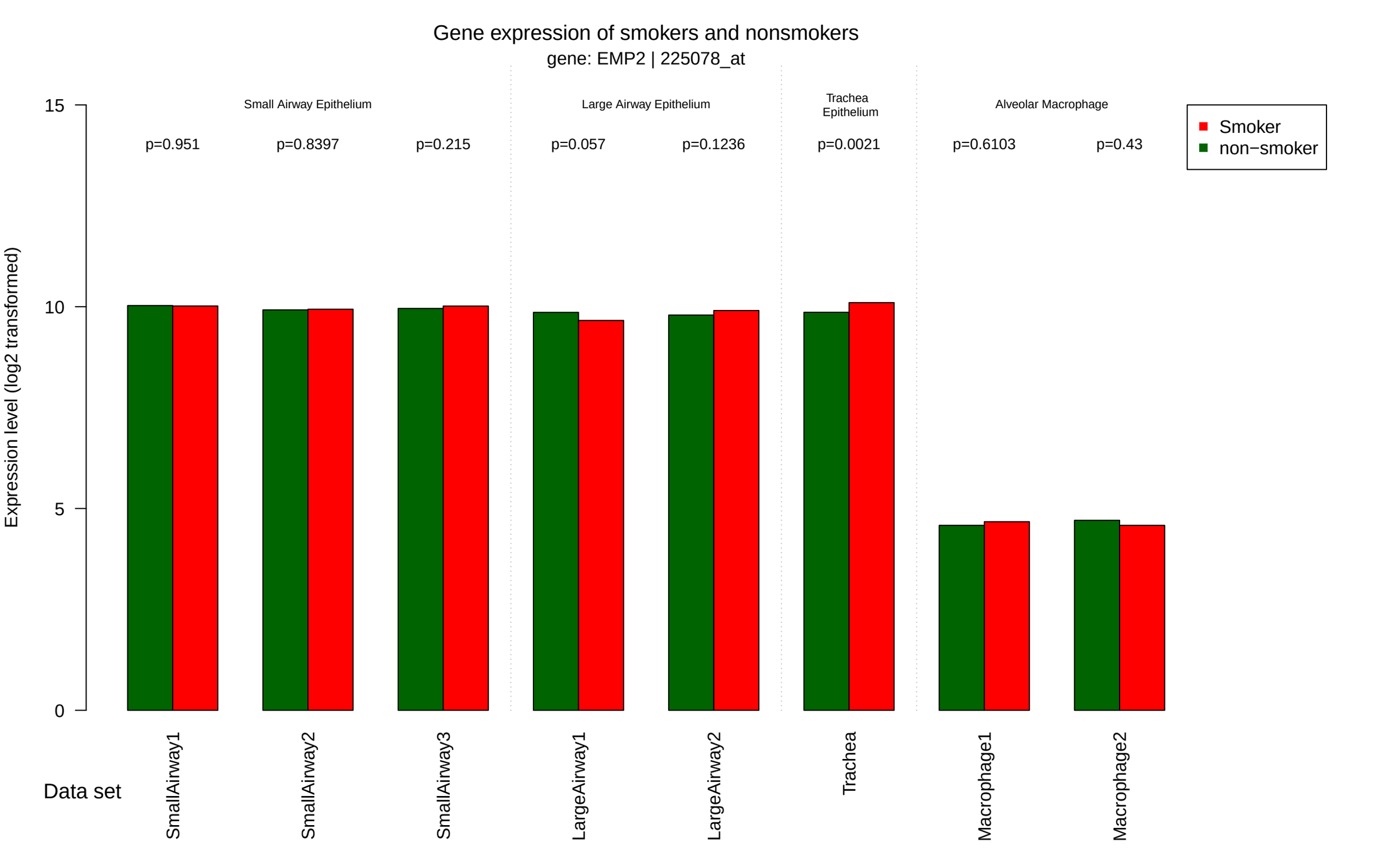
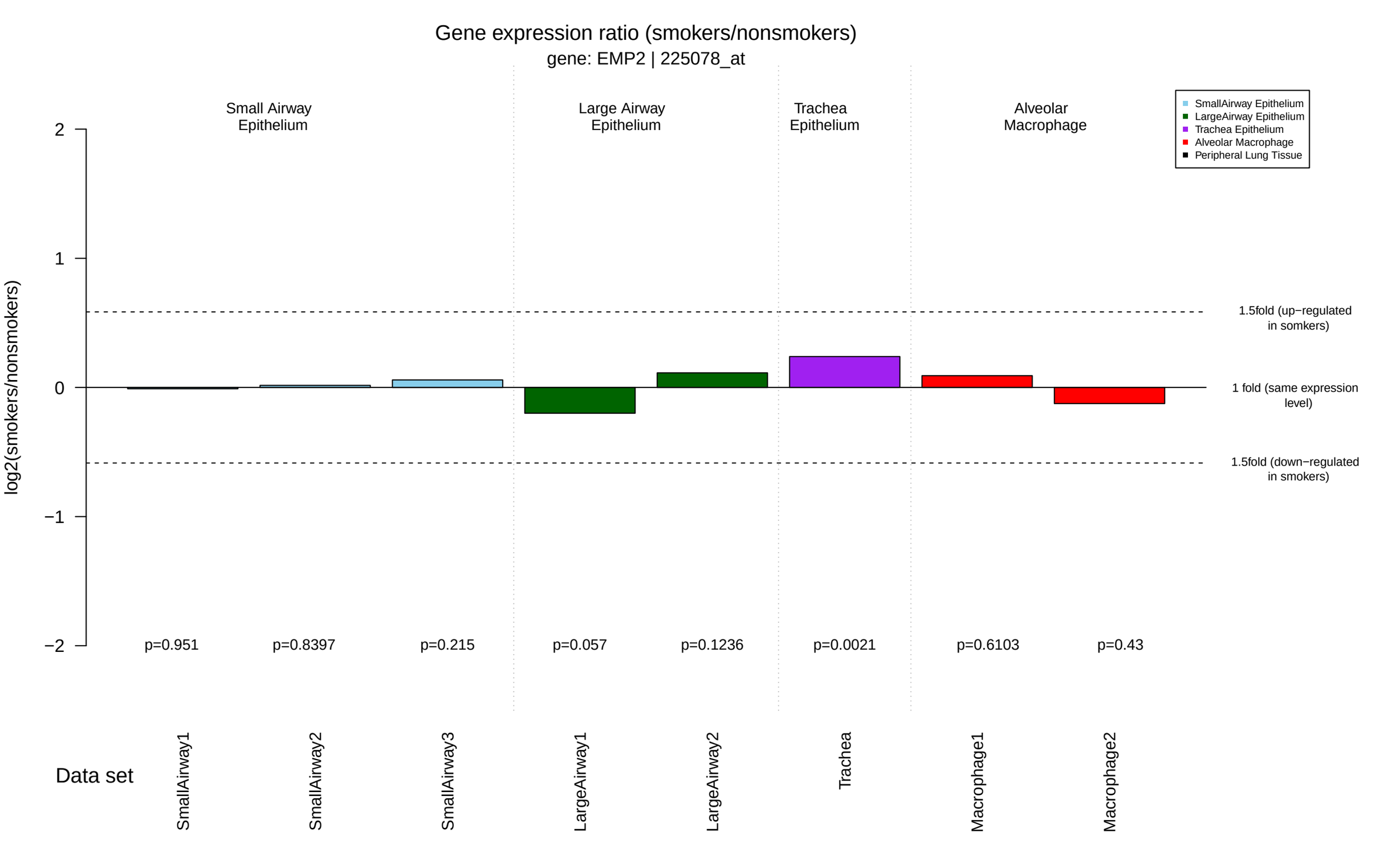
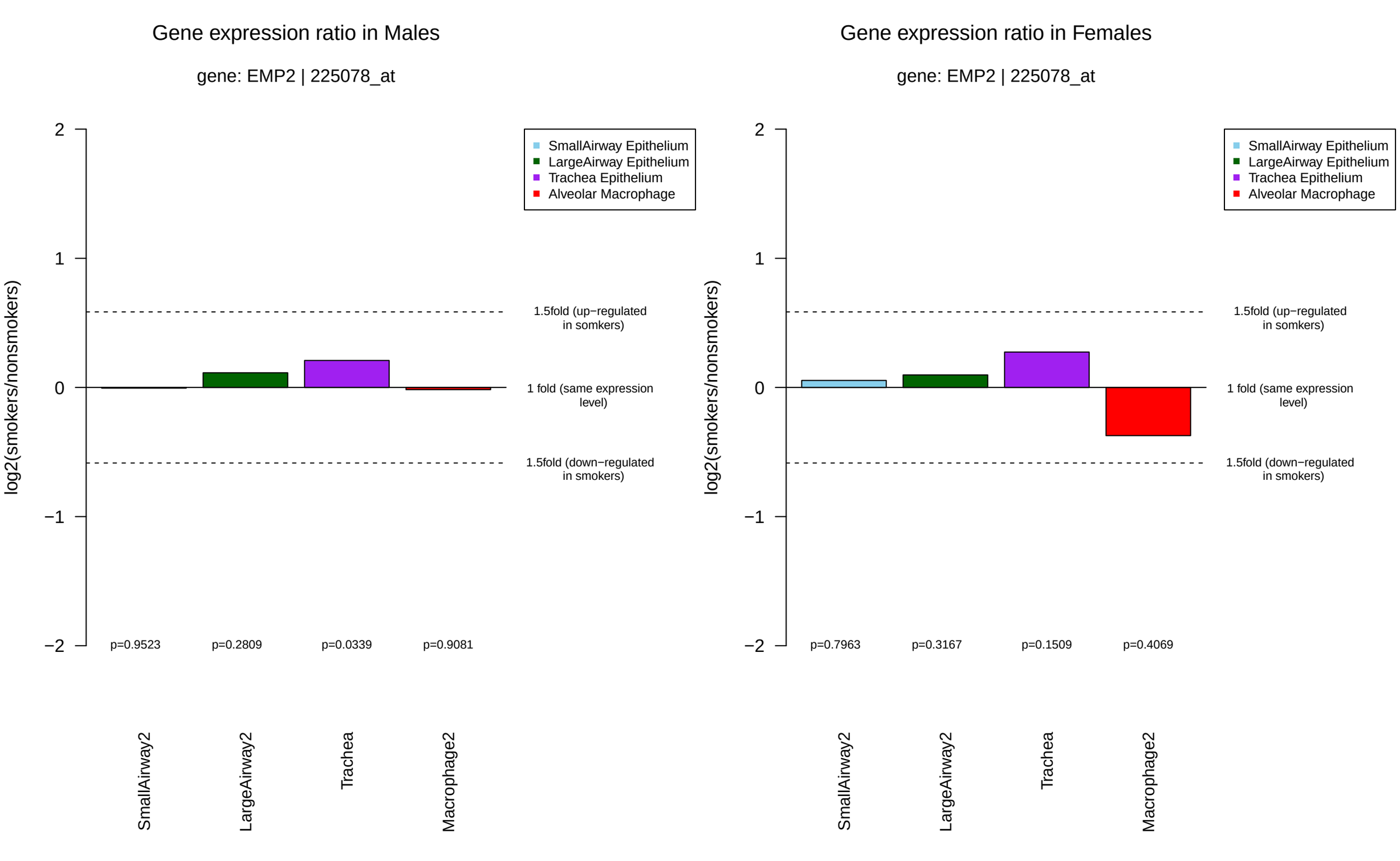
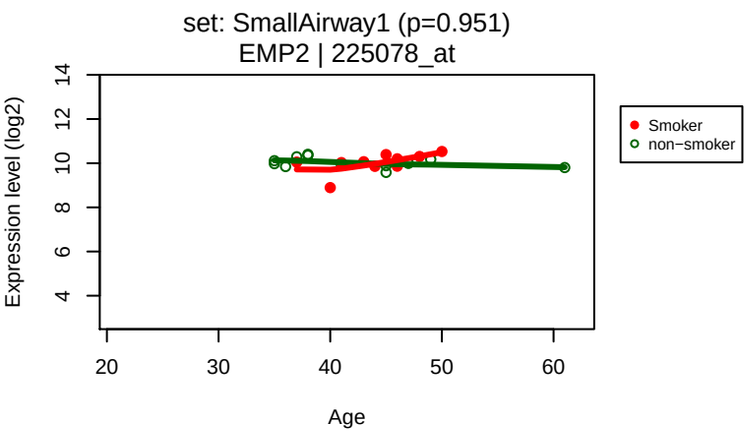
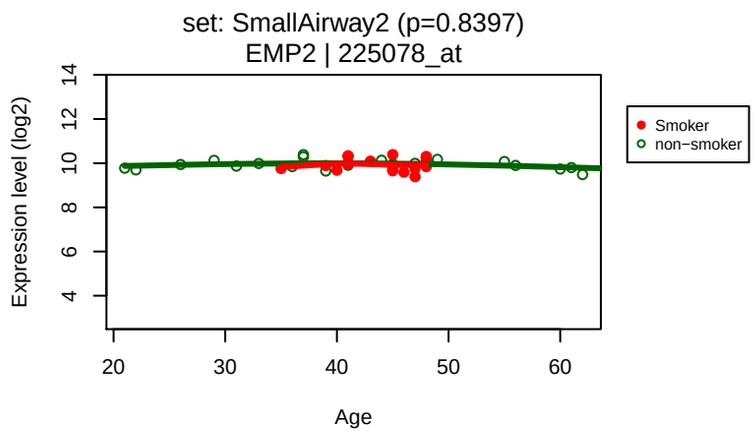
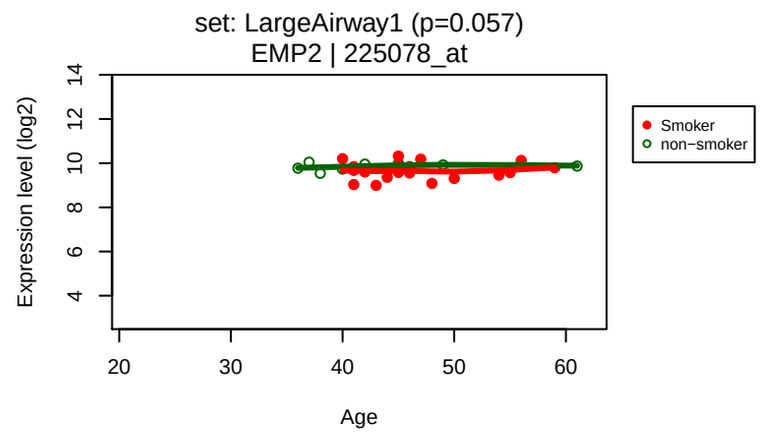

Gene expression of smokers and nonsmokers
gene: EMP2 | 225078_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.951
p=0.8397
p=0.215
p=0.057
p=0.1236
p=0.0021
p=0.6103
p=0.43
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

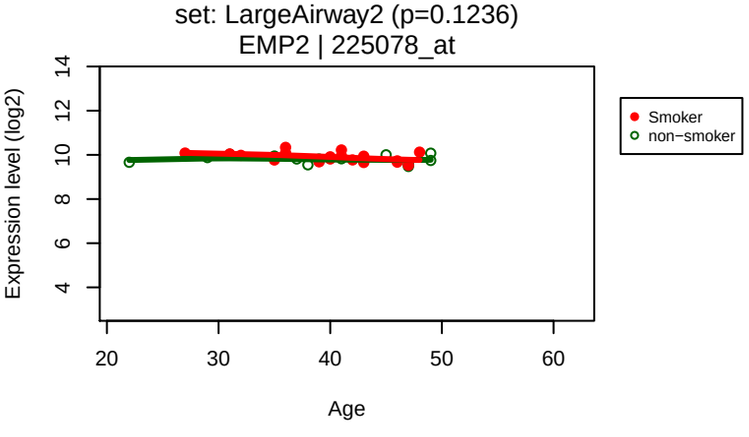

Gene expression ratio (smokers/nonsmokers)
gene: EMP2 | 225078_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.951
p=0.8397
p=0.215
p=0.057
p=0.1236
p=0.0021
p=0.6103
p=0.43
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

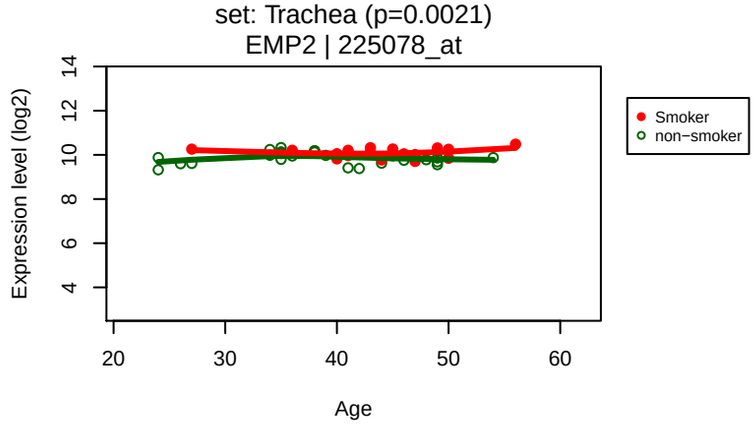

Gene expression ratio in Males
gene: EMP2 | 225078_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9523
p=0.2809
p=0.0339
p=0.9081
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EMP2 | 225078_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7963
p=0.3167
p=0.1509
p=0.4069
SmallAirway2
LargeAirway2
Trachea
Macrophage2

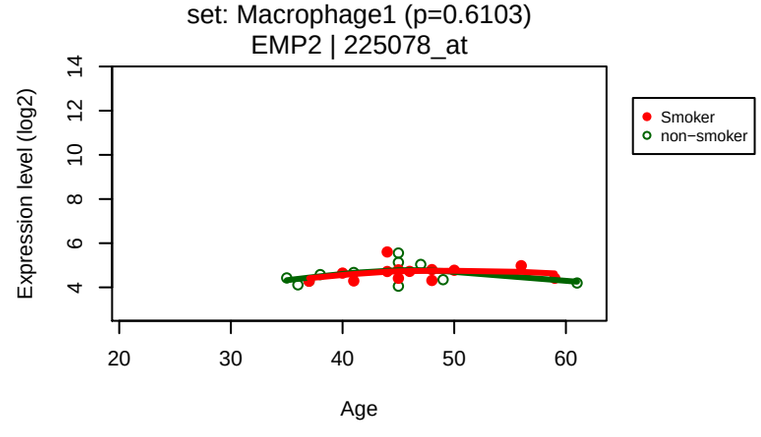

set: SmallAirway1 (p=0.951)
EMP2 | 225078_at
Expression level (log2)
Age
Smoker
non−smoker

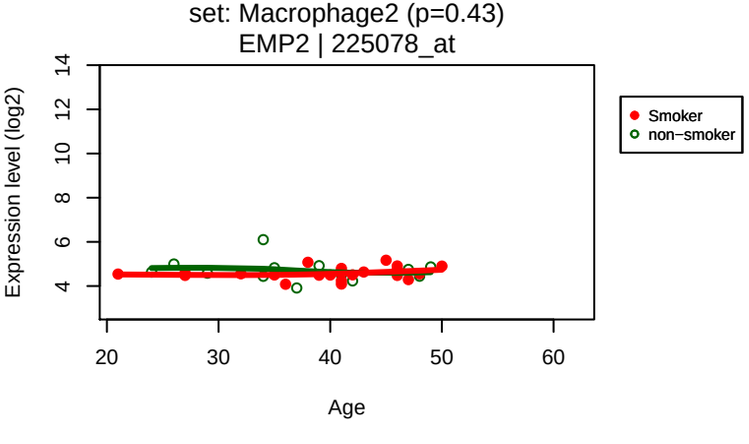

set: SmallAirway2 (p=0.8397)
EMP2 | 225078_at
Expression level (log2)
Age
Smoker
non−smoker

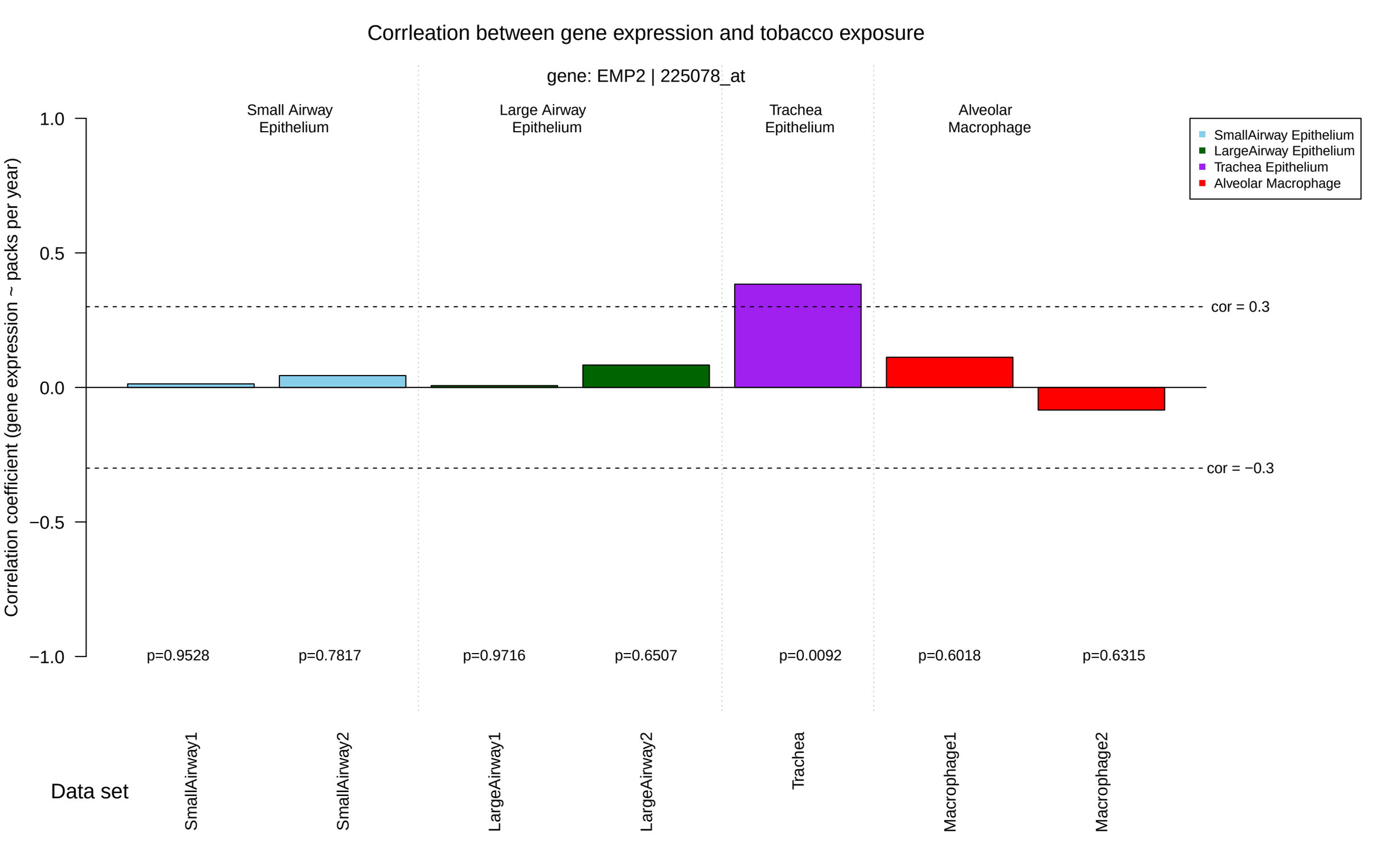

set: LargeAirway1 (p=0.057)
EMP2 | 225078_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1236)
EMP2 | 225078_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0021)
EMP2 | 225078_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6103)
EMP2 | 225078_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.43)
EMP2 | 225078_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EMP2 | 225078_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9528
p=0.7817
p=0.9716
p=0.6507
p=0.0092
p=0.6018
p=0.6315
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2